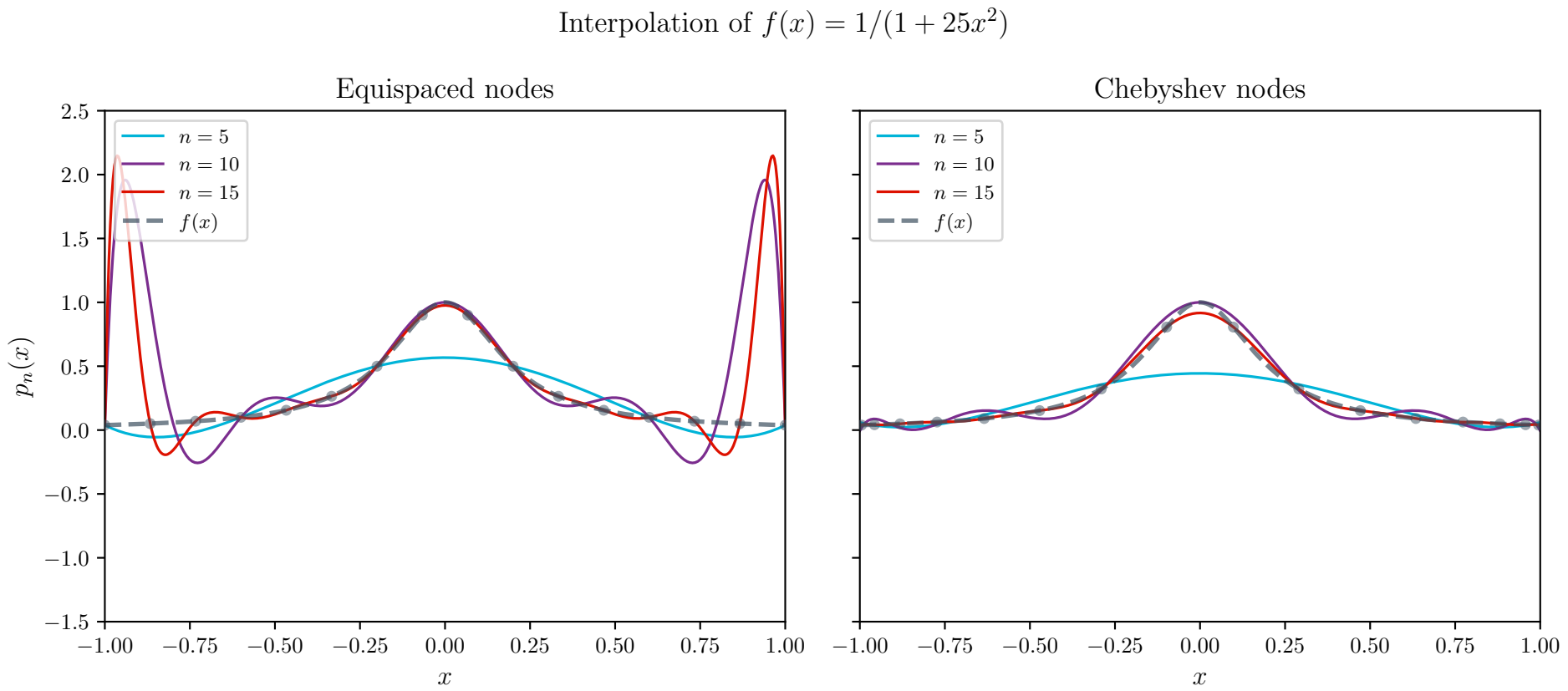
Interpolation of $f(x) = 1/(1 + 25x^2)$
Equispaced nodes
Chebyshev nodes
$n = 5$
$n = 10$
$n = 15$
$f(x)$
$p_n(x)$
x
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00